

## Informe taller software

**Presentado por:**

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En este taller estamos configurando una API en Laravel para evaluar su comportamiento cuando maneja grandes volúmenes de datos y peticiones concurrentes.

Para iniciar el taller, se verificó la disponibilidad de PHP con una versión superior a 8.0 y Composer en el equipo local. Seguido de esto se configuró la base de datos siguiendo estos pasos:

Creación de la base de datos “tallersoftware”, se ajustaron las variables de conexión con el archivo .env, después se llevó a cabo las migraciones y el llenado de datos de prueba.

Con los datos listos se inició el servidor de desarrollo local

php artisan serve --host=0.0.0.0 --port=8000

La URL base del servicio quedó establecida en <http://localhost:8000/api>

Para verificar el comportamiento de los endpoints sin depender únicamente de la consola, se utilizó la extensión REST Client en Visual Studio Code mediante un archivo pruebas.http, además de pruebas directas en el navegador:

GET <http://localhost:8000/api/users/emails>

GET <http://localhost:8000/api/users/over-twenty>

GET <http://localhost:8000/api/users>

POST <http://localhost:8000/api/users/bulk>

Hallazgo y análisis de falla: Error 500 y agotamiento de memoria

Al probar el endpoint GET /api/users/emails (y en escenarios con volumen completo de datos), el servicio no entregó los registros y retornó un código de estado HTTP 500 Internal Server Error.

Revisión del Log (storage/logs/laravel.log)

Al inspeccionar el final del archivo storage/logs/laravel.log, se localizó el siguiente registro de error crítico:

```
[2026-10-08 16:51:50] local.ERROR: Allowed memory size of 536870912 bytes exhausted (tried to allocate 20480 bytes) {"exception":"[object] (Symfony\\[stacktrace]
#0 {main}
")
```

Este error se dio porque el endpoint ejecuta una consulta que intenta cargar todos los registros existentes en memoria simultáneamente, sin ningún tipo de límite o partición este, límite de PHP fue superado ya que el entorno tiene un tope de memoria configurado en 512 MB (536,870,912 bytes). Al sobrepasar, el proceso se interrumpe inmediatamente y devuelve error 500.

Para solucionar la caída del servidor y permitir que las pruebas de carga midan tiempos representativos, se aplicó la recomendación del taller: refactorizar el controlador de usuarios (UserController.php).

Cambios realizados en el código:

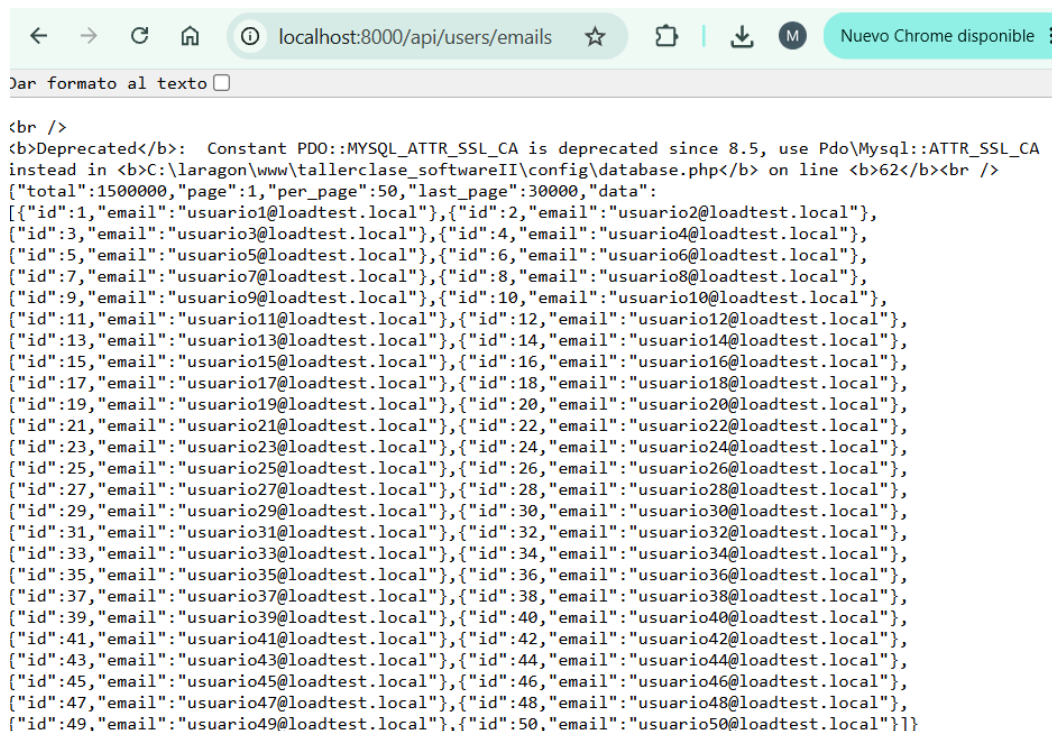
Se eliminaron las consultas directas sin tope (all() o get()).

Se incorporaron parámetros de consulta (page y per\_page), fijando un límite controlado por página.

La respuesta ahora entrega los datos fragmentados junto con la estructura de metadatos de paginación (total, page, per\_page, last\_page, data).

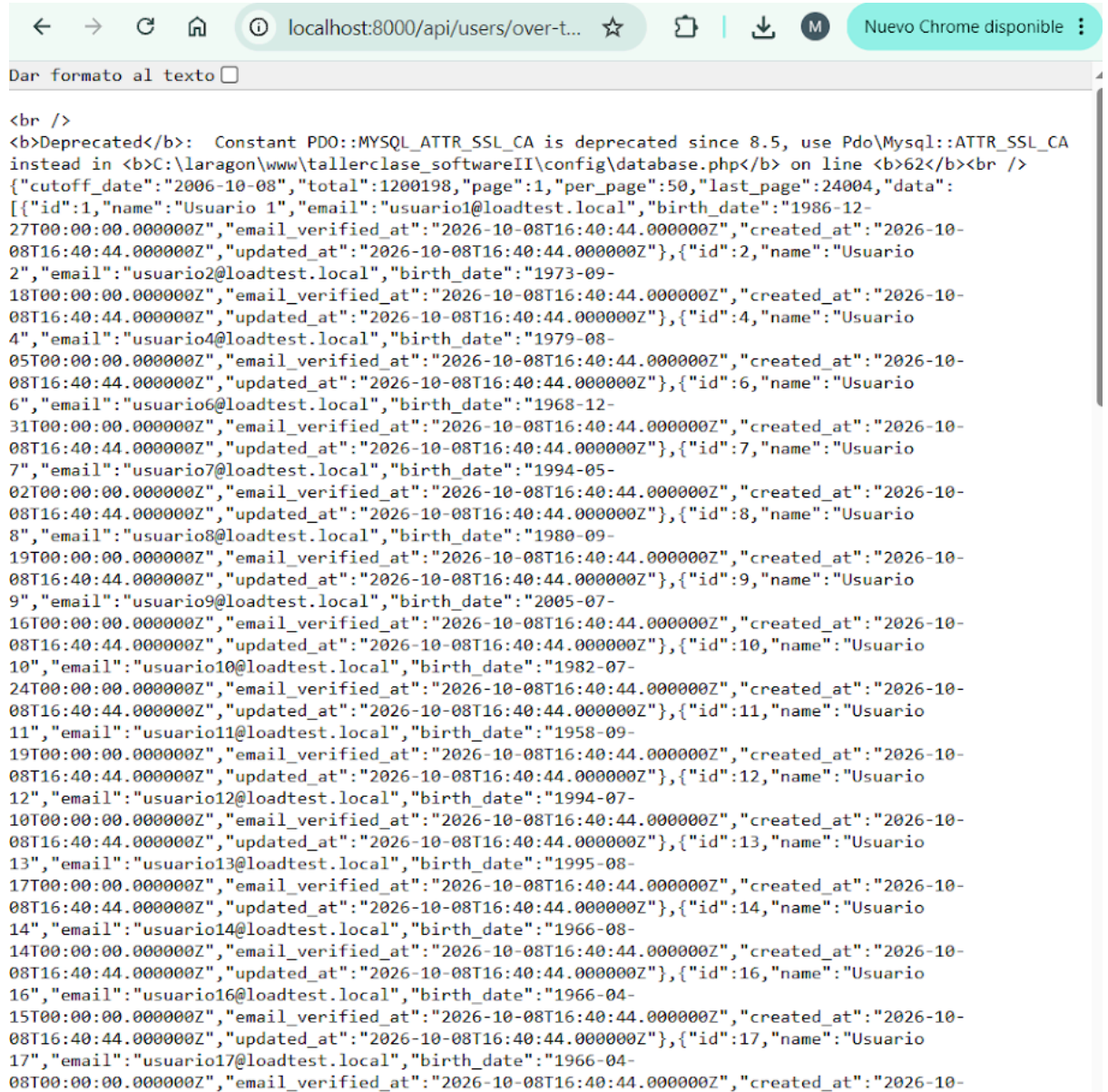
Después procedimos a realizar la revalidaron las peticiones en el navegador y mediante REST Client, confirmando que la API procesa el dataset completo de 1.500.000 registros sin saturar la memoria:

- Endpoint de correos (/api/users/emails): Devuelve exitosamente los primeros 50 registros de correos de un total de 1.500.000 usuarios.



```
<br />
<b>Deprecated</b>: Constant PDO::MYSQL_ATTR_SSL_CA is deprecated since 8.5, use Pdo\Mysql::ATTR_SSL_CA
instead in <b>C:\laragon\www\tallerclase_softwareII\config\database.php</b> on line <b>62</b><br />
{"total":1500000,"page":1,"per_page":50,"last_page":30000,"data":
[{"id":1,"email":"usuario1@loadtest.local"}, {"id":2,"email":"usuario2@loadtest.local"},
{"id":3,"email":"usuario3@loadtest.local"}, {"id":4,"email":"usuario4@loadtest.local"},
{"id":5,"email":"usuario5@loadtest.local"}, {"id":6,"email":"usuario6@loadtest.local"},
{"id":7,"email":"usuario7@loadtest.local"}, {"id":8,"email":"usuario8@loadtest.local"},
{"id":9,"email":"usuario9@loadtest.local"}, {"id":10,"email":"usuario10@loadtest.local"},
{"id":11,"email":"usuario11@loadtest.local"}, {"id":12,"email":"usuario12@loadtest.local"},
{"id":13,"email":"usuario13@loadtest.local"}, {"id":14,"email":"usuario14@loadtest.local"},
{"id":15,"email":"usuario15@loadtest.local"}, {"id":16,"email":"usuario16@loadtest.local"},
{"id":17,"email":"usuario17@loadtest.local"}, {"id":18,"email":"usuario18@loadtest.local"},
{"id":19,"email":"usuario19@loadtest.local"}, {"id":20,"email":"usuario20@loadtest.local"},
{"id":21,"email":"usuario21@loadtest.local"}, {"id":22,"email":"usuario22@loadtest.local"},
{"id":23,"email":"usuario23@loadtest.local"}, {"id":24,"email":"usuario24@loadtest.local"},
{"id":25,"email":"usuario25@loadtest.local"}, {"id":26,"email":"usuario26@loadtest.local"},
{"id":27,"email":"usuario27@loadtest.local"}, {"id":28,"email":"usuario28@loadtest.local"},
{"id":29,"email":"usuario29@loadtest.local"}, {"id":30,"email":"usuario30@loadtest.local"},
{"id":31,"email":"usuario31@loadtest.local"}, {"id":32,"email":"usuario32@loadtest.local"},
{"id":33,"email":"usuario33@loadtest.local"}, {"id":34,"email":"usuario34@loadtest.local"},
{"id":35,"email":"usuario35@loadtest.local"}, {"id":36,"email":"usuario36@loadtest.local"},
{"id":37,"email":"usuario37@loadtest.local"}, {"id":38,"email":"usuario38@loadtest.local"},
{"id":39,"email":"usuario39@loadtest.local"}, {"id":40,"email":"usuario40@loadtest.local"},
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{"id":45,"email":"usuario45@loadtest.local"}, {"id":46,"email":"usuario46@loadtest.local"},
{"id":47,"email":"usuario47@loadtest.local"}, {"id":48,"email":"usuario48@loadtest.local"},
{"id":49,"email":"usuario49@loadtest.local"}, {"id":50,"email":"usuario50@loadtest.local"}]]
```

- endpoint con filtro de edad (/api/users/over-twenty):Calcula la fecha de corte (cutoff\_date), contabiliza 1.200.198 usuarios mayores de 20 años y entrega el bloque paginado sin colapsar.



```

<br />
<b>Deprecated</b>: Constant PDO::MYSQL_ATTR_SSL_CA is deprecated since 8.5, use PDO::MYSQL_ATTR_SSL_CA
instead in <b>C:\laragon\www\tallerclase_softwareII\config\database.php</b> on line <b>62</b><br />
{"cutoff_date":"2006-10-08","total":1200198,"page":1,"per_page":50,"last_page":24004,"data":
[{"id":1,"name":"Usuario 1","email":"usuario1@loadtest.local","birth_date":"1986-12-
27T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"}, {"id":2,"name":"Usuario
2","email":"usuario2@loadtest.local","birth_date":"1973-09-
18T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"}, {"id":4,"name":"Usuario
4","email":"usuario4@loadtest.local","birth_date":"1979-08-
05T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"}, {"id":6,"name":"Usuario
6","email":"usuario6@loadtest.local","birth_date":"1968-12-
31T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"}, {"id":7,"name":"Usuario
7","email":"usuario7@loadtest.local","birth_date":"1994-05-
02T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"}, {"id":8,"name":"Usuario
8","email":"usuario8@loadtest.local","birth_date":"1980-09-
19T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"}, {"id":9,"name":"Usuario
9","email":"usuario9@loadtest.local","birth_date":"2005-07-
16T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"}, {"id":10,"name":"Usuario
10","email":"usuario10@loadtest.local","birth_date":"1982-07-
24T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"}, {"id":11,"name":"Usuario
11","email":"usuario11@loadtest.local","birth_date":"1958-09-
19T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"}, {"id":12,"name":"Usuario
12","email":"usuario12@loadtest.local","birth_date":"1994-07-
10T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"}, {"id":13,"name":"Usuario
13","email":"usuario13@loadtest.local","birth_date":"1995-08-
17T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"}, {"id":14,"name":"Usuario
14","email":"usuario14@loadtest.local","birth_date":"1966-08-
14T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"}, {"id":16,"name":"Usuario
16","email":"usuario16@loadtest.local","birth_date":"1966-04-
15T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"}, {"id":17,"name":"Usuario
17","email":"usuario17@loadtest.local","birth_date":"1966-04-
08T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-

```

- Endpoint general de usuarios (/api/users):Responde con la estructura por defecto (per\_page: 50 y un total de 30.000 páginas).

```
localhost:8000/api/users

Dar formato al texto

<br />
<b>Deprecated</b>: Constant PDO::MYSQL_ATTR_SSL_CA is deprecated since 8.5, use Pdo\Mysql::ATTR_SSL_CA
instead in <b>C:\laragon\www\tallerclase_softwareII\config\database.php</b> on line <b>62</b><br />
{"total":150000,"page":1,"per_page":50,"last_page":3000,"data":[{"id":1,"name":"Usuario
1","email":"usuario1@loadtest.local","birth_date":"1986-12-
27T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"},{"id":2,"name":"Usuario
2","email":"usuario2@loadtest.local","birth_date":"1973-09-
18T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"},{"id":3,"name":"Usuario
3","email":"usuario3@loadtest.local","birth_date":"2008-04-
29T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"},{"id":4,"name":"Usuario
4","email":"usuario4@loadtest.local","birth_date":"1979-08-
05T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"},{"id":5,"name":"Usuario
5","email":"usuario5@loadtest.local","birth_date":"2015-10-
05T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"},{"id":6,"name":"Usuario
6","email":"usuario6@loadtest.local","birth_date":"1968-12-
31T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"},{"id":7,"name":"Usuario
7","email":"usuario7@loadtest.local","birth_date":"1994-05-
02T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"},{"id":8,"name":"Usuario
8","email":"usuario8@loadtest.local","birth_date":"1980-09-
19T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"},{"id":9,"name":"Usuario
9","email":"usuario9@loadtest.local","birth_date":"2005-07-
16T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"},{"id":10,"name":"Usuario
10","email":"usuario10@loadtest.local","birth_date":"1982-07-
24T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"},{"id":11,"name":"Usuario
11","email":"usuario11@loadtest.local","birth_date":"1958-09-
19T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"},{"id":12,"name":"Usuario
12","email":"usuario12@loadtest.local","birth_date":"1994-07-
10T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"},{"id":13,"name":"Usuario
13","email":"usuario13@loadtest.local","birth_date":"1995-08-
17T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"},{"id":14,"name":"Usuario
14","email":"usuario14@loadtest.local","birth_date":"1966-08-
14T00:00:00.000000Z","email_verified_at":"2026-10-08T16:40:44.000000Z","created_at":"2026-10-
08T16:40:44.000000Z","updated_at":"2026-10-08T16:40:44.000000Z"},{"id":15,"name":"Usuario
15","email":"usuario15@loadtest.local","birth_date":"2013-11-
```

- Endpoint con parámetro personalizado (/api/users?per\_page=...): Permite personalizar la cantidad de resultados por bloque (en este caso solicitando hasta 100 registros), respondiendo de forma fluida

Prueba realizada para comprobar el buen funcionamiento.

## Prueba de carga

Prueba para 40 usuarios con un tiempo de 2 ramp up

Con una carga baja de 5 usuarios concurrentes, la API se comportó de forma estable, logrando un 0% de fallos y atendiendo todas las peticiones con éxito. Esto confirma que la paginación que implementamos funciona bien y no satura la memoria del servidor bajo un tráfico ligero



LOCUST

Host

http://localhost:8000/api

Status

RUNNING

Users

40

RPS

1.4

Failures

0%

EDIT

STOP

RESET



STATISTICS

CHARTS

FAILURES

EXCEPTIONS

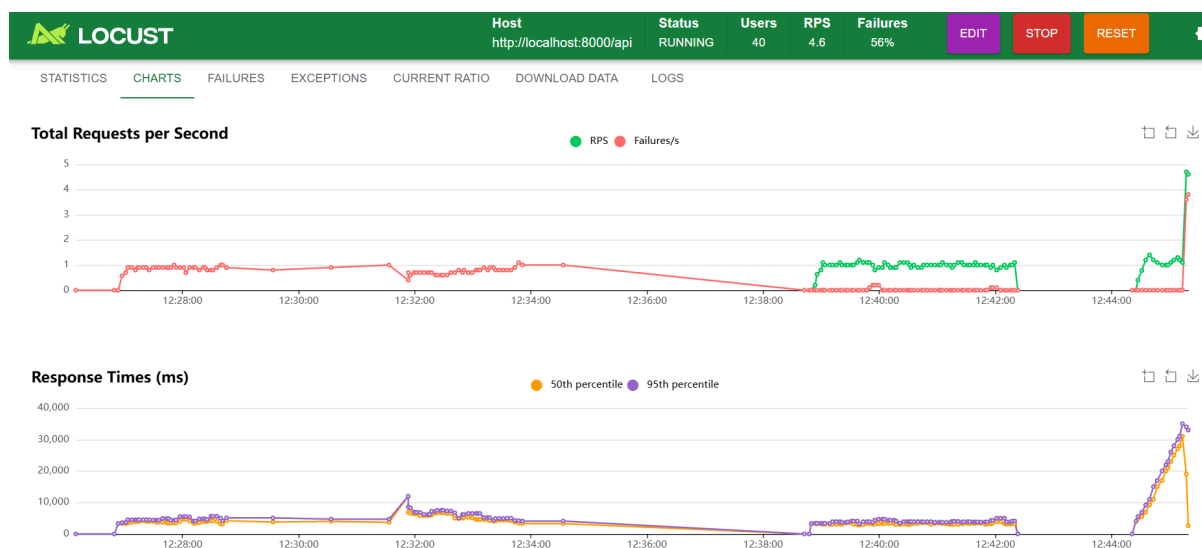
CURRENT RATIO

DOWNLOAD DATA

LOGS



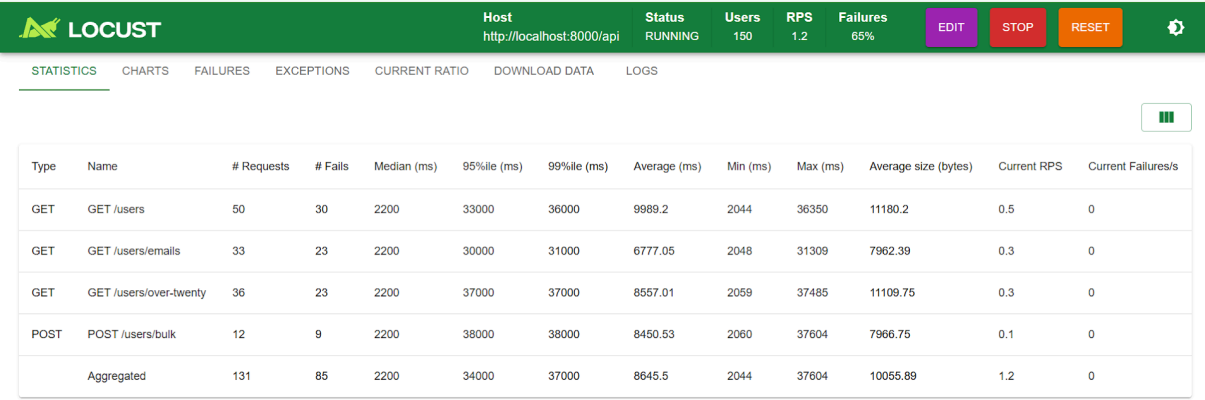
| Type | Name                   | # Requests | # Fails | Median (ms) | 95%ile (ms) | 99%ile (ms) | Average (ms) | Min (ms) | Max (ms) | Average size (bytes) | Current RPS | Current Failures/s |
|------|------------------------|------------|---------|-------------|-------------|-------------|--------------|----------|----------|----------------------|-------------|--------------------|
| GET  | GET /users             | 10         | 0       | 6000        | 11000       | 11000       | 7163.74      | 3141     | 11177    | 12451                | 0.3         | 0                  |
| GET  | GET /users/emails      | 4          | 0       | 6100        | 8100        | 8100        | 6507.06      | 4941     | 8105     | 2510                 | 0.4         | 0                  |
| GET  | GET /users/over-twenty | 6          | 0       | 6500        | 9200        | 9200        | 7145.74      | 5109     | 9157     | 12484                | 0.6         | 0                  |
| POST | POST /users/bulk       | 1          | 0       | 9403.21     | 9400        | 9400        | 9403.21      | 9403     | 9403     | 868                  | 0.1         | 0                  |
|      | Aggregated             | 21         | 0       | 6900        | 10000       | 11000       | 7140.16      | 3141     | 11177    | 10015.33             | 1.4         | 0                  |



## Prueba de estrés

Prueba para 150 usuario con un tiempo de 10 ramp up

Al correr la prueba en Locust con los 150 usuarios nos dimos cuenta de que el sistema básicamente no aguantó. Aunque logramos solucionar el problema inicial de la memoria paginando las respuestas para que no saliera el error 500 de una vez, meterle tanta gente al tiempo hizo que la gráfica se pusiera en rojo con un 65% de fallos



### Segunda prueba de estrés

Al subir la carga a 300 usuarios concurrentes, el servidor local colapsó en su totalidad, registrando una tasa de fallas del 96% y tiempos de respuesta de más de 40 segundos.

